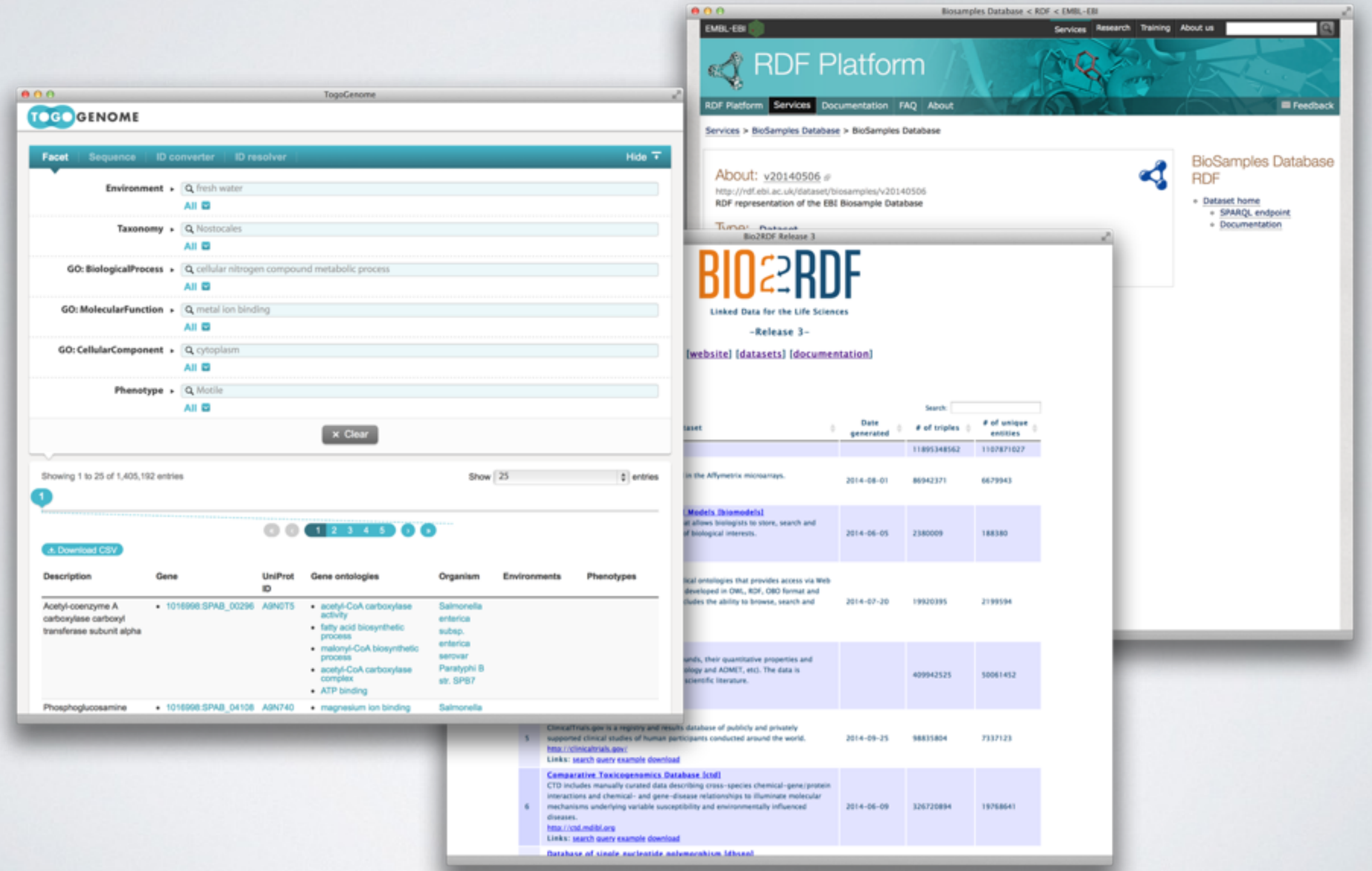
A SPARQL Endpoint Profiler for Efficient Question Answering Systems

Yasunori Yamamoto
Database Center for Life Science (DBCLS)



Various SPARQL Endpoints



SPARQL

SPARQL Protocol and RDF Query Language



You can get any data necessarily and sufficiently from them…?

A SPARQL editor you might often see…

Virtuoso SPARQL Query Editor

About | Namespace Prefixes | Inference rules

Default Data Set Name (Graph IRI)

Query Text

```
select distinct ?Concept where {[] a ?Concept} LIMIT 100
```

Sponging: Use only local data (including data re

Results Format: HTML

Execution timeout: 0 millisecond

Options: Strict checking of void variables

(The result can only be sent back to browser, not saved on the server, see details)

Run Query Reset

Virtuoso … r Edi…

Almost nothing here…



dean beyett

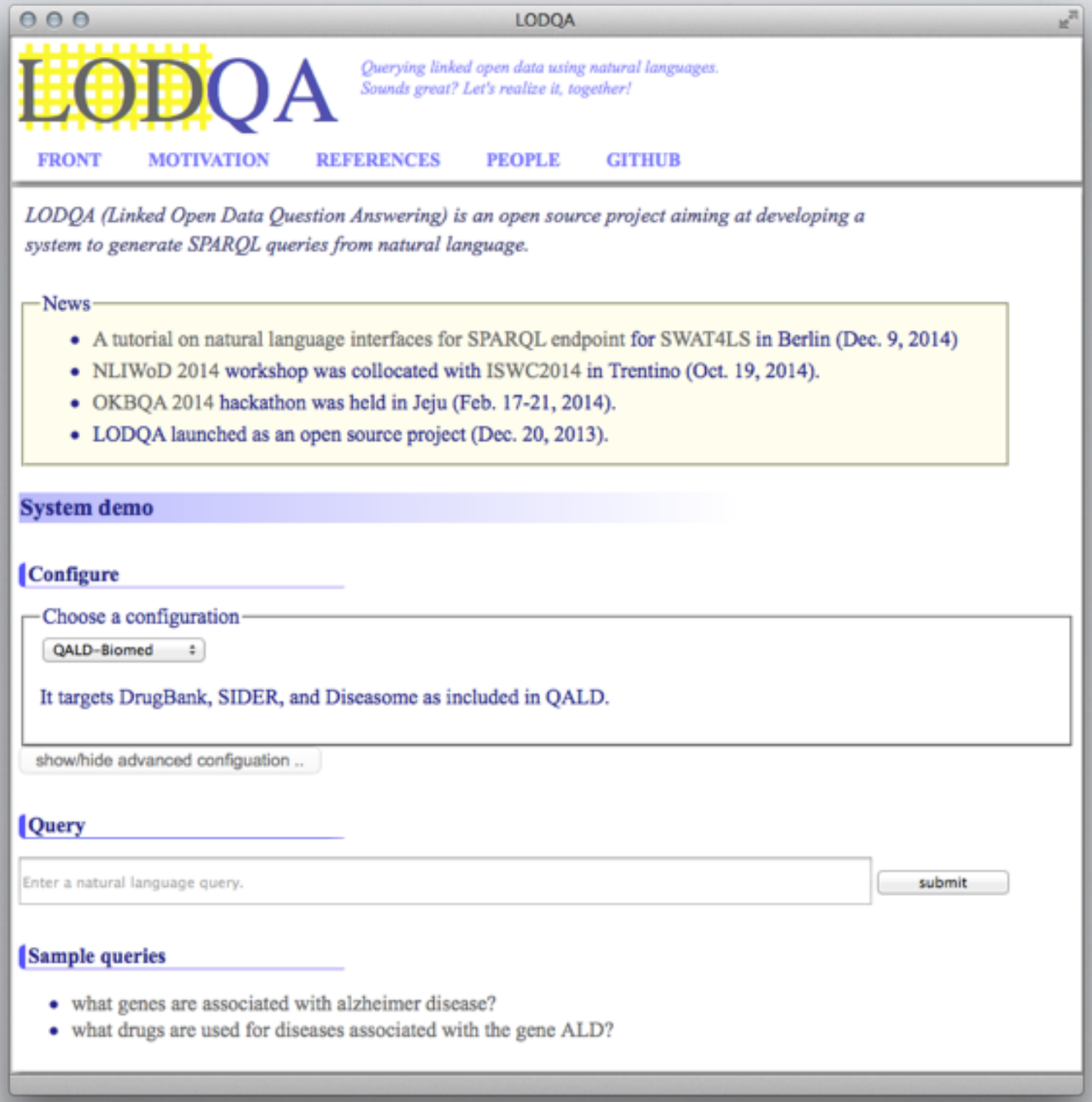
Use the force

We’d like to make this happen.



Lower Columbia College (LCC)

Question Answering



"What genes are associated with Alzheimer disease?"

```
SELECT ?t1
    WHERE {
   ?t1 [:isa] [genes] .
   ?t2 [:isa] [alzheimer disease] .
   ?t1 [associated with] ?t2 .
}
```

Mapping to existing vocabularies

[genes] = a [gene] class
[alzheimer disease] = instance of a [disease] class

Out of this study's focus

PubDictionary / identifiers.org

Example: BIO2RDF

[gene] Class = bio2rdf:omim_vocabulary:Gene

[disease] Class = bio2rdf:umls_vocabulary:Resource

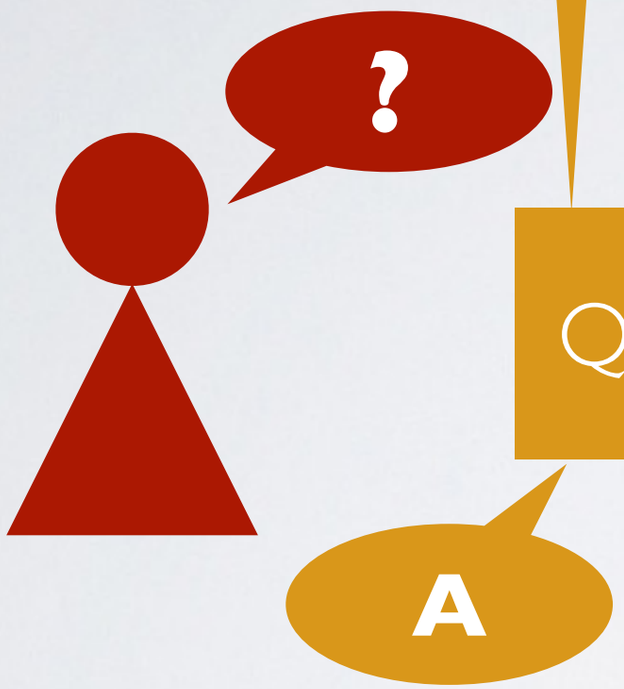
[disease] Class = bio2rdf:omim_vocabulary:Phenotype

A QA system wants to know if there is any path between them.

Naive method

Finding EPs that can compose a path between a [gene] class and [alzheimer disease] .

gene/omim/umls?



SPARQL Endpoint

SPARQL Endpoint

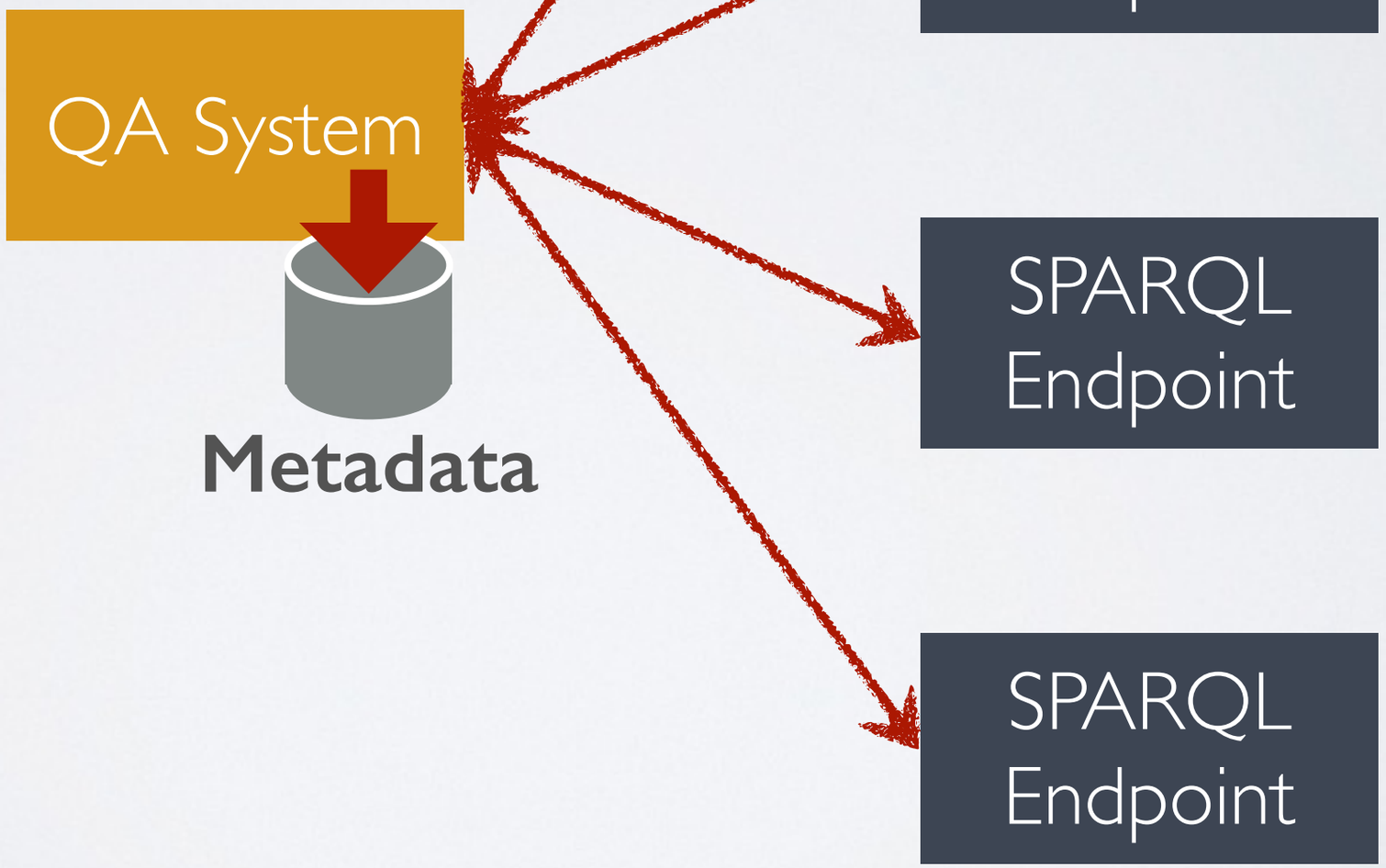
SPARQL Endpoint

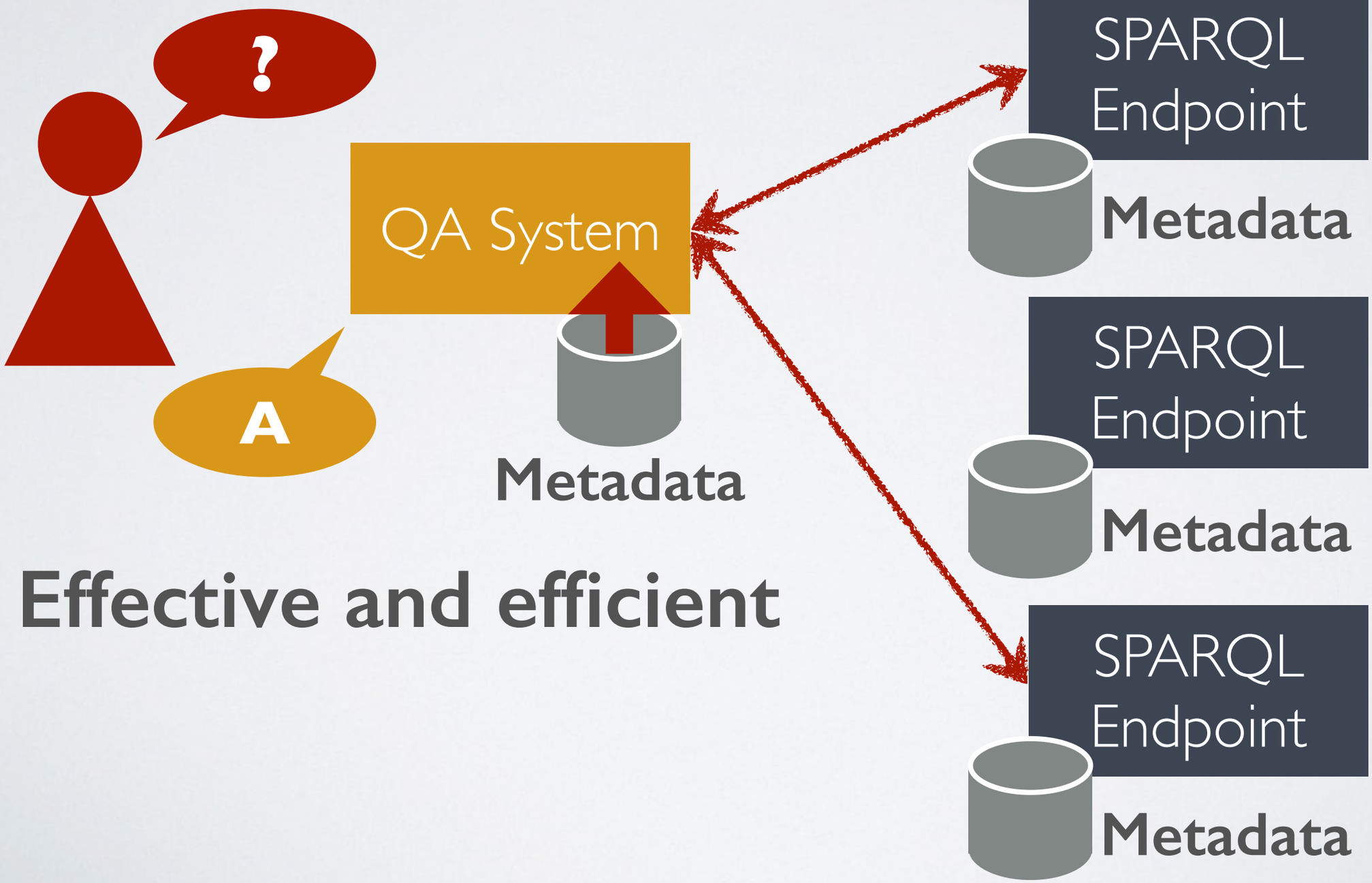
SPARQL Endpoint

Slow and inefficient

Our approach

Previously gathers metadata of each endpoint

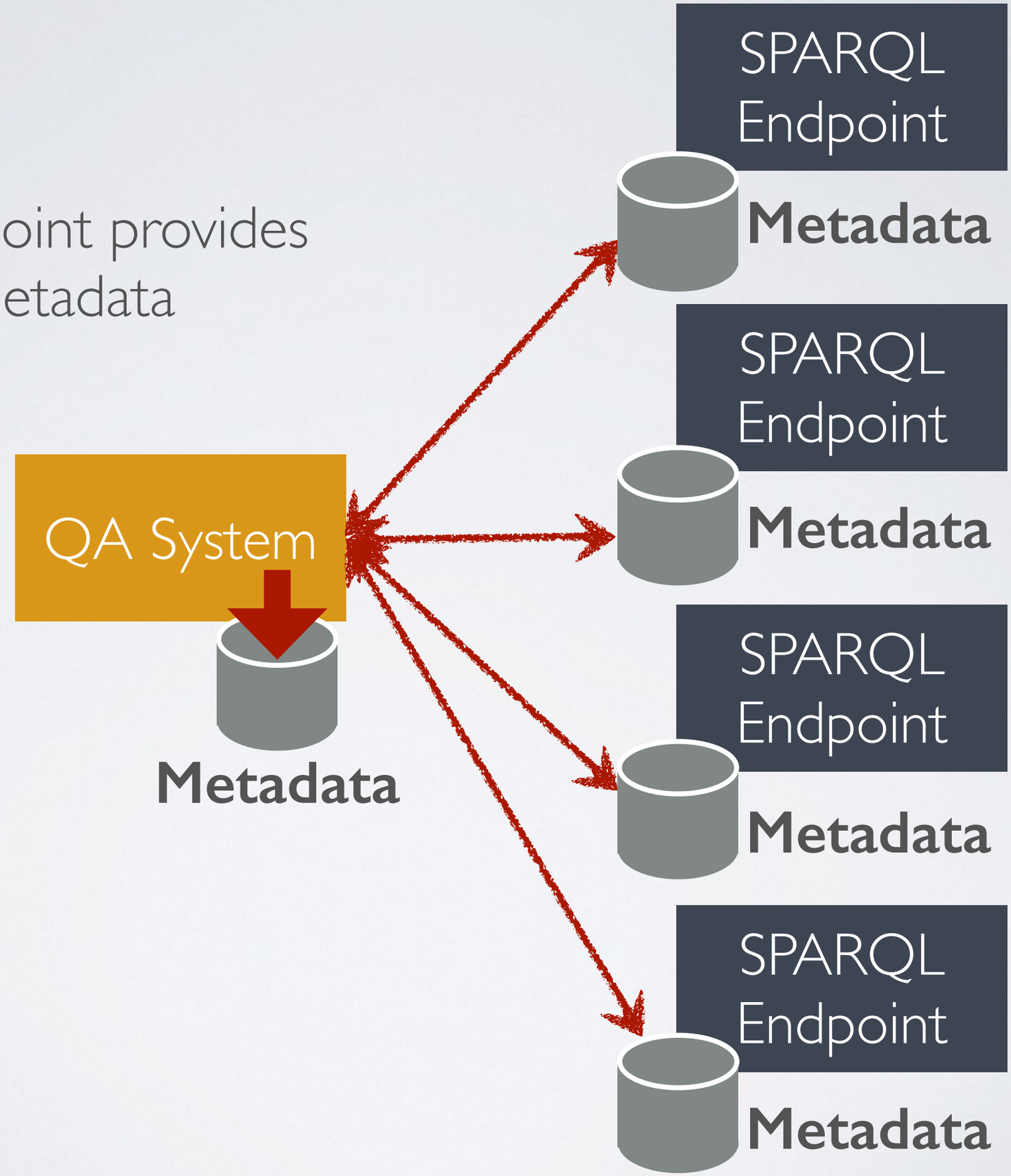




Effective and efficient

Ideal way

Each endpoint provides its metadata



VoID

SD

Yes, they're fine!

Vocabulary of Interlinked Datasets

SPARQL 1.1 Service Description

But, we cannot get…

Data about relationships between classes

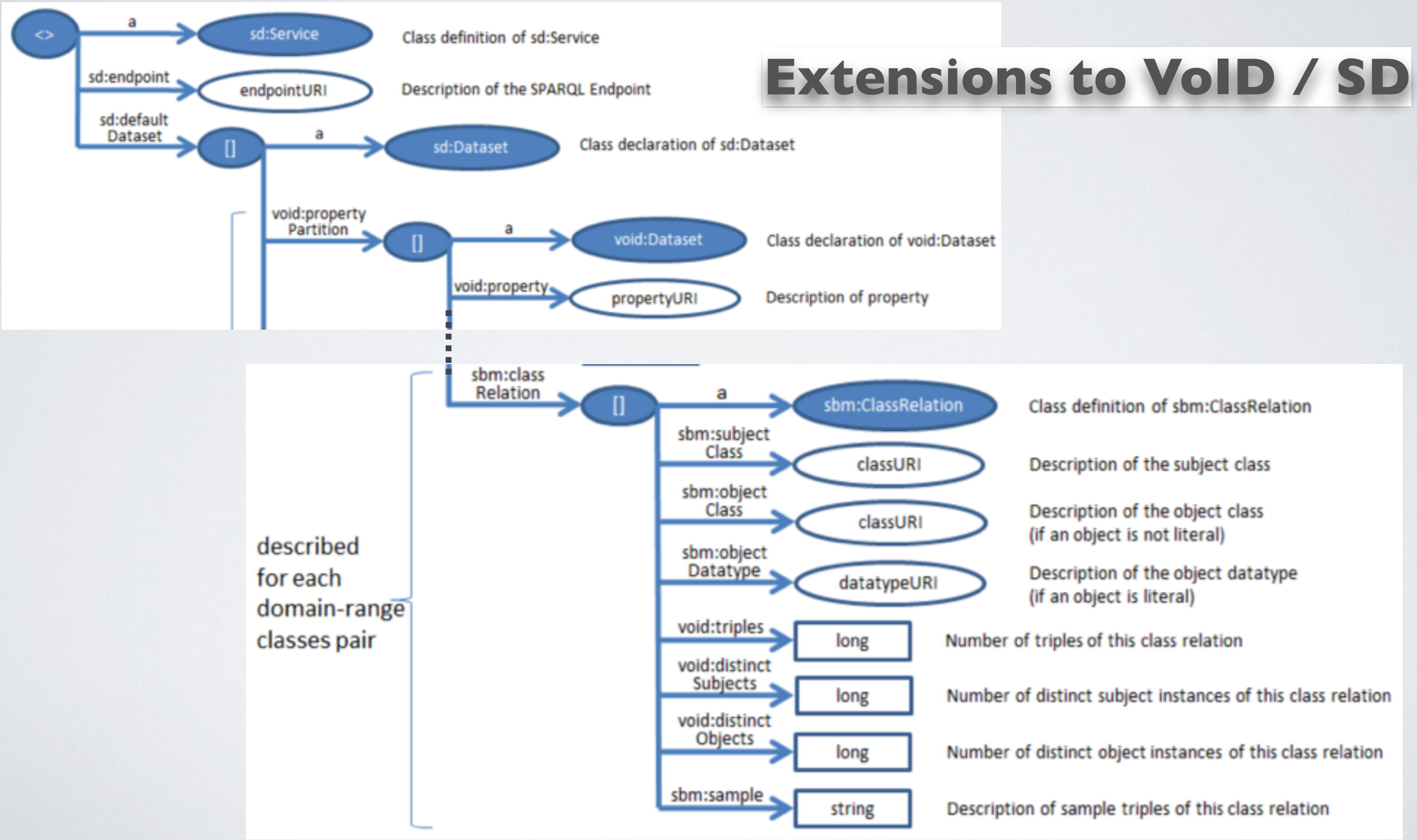


Cap9
APCS
PRNP
:

EEC SYNDROME 1
TOXEMIA OF PREGNANCY
DIABETES
:

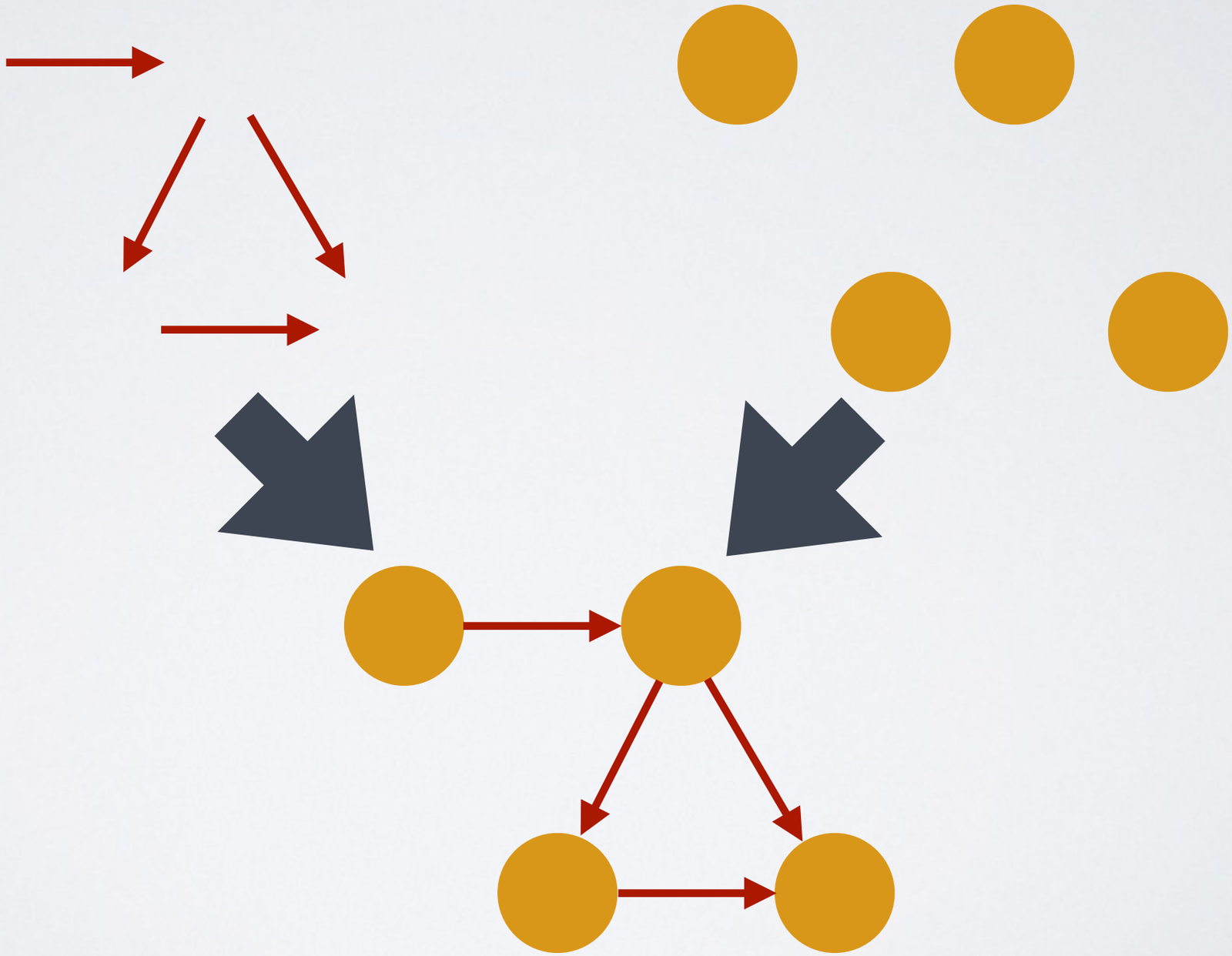
SPARQL Builder Metadata

Extensions to VoID / SD



Obtain all the relationships

Predicate first vs. Class first



http://stats.lod2.eu/stats

Criterion	Average	Min	Max	**Median**	Total
Classes	1,976.29	1	50,303	**5.0**	509,884
Properties	33.48	1	885	**21.0**	9,608

Obtain metadata using SPARQL

A locally declared class

any resource that appears as an object of **rdf:type** in the dataset.

```
:sycamore rdf:type :Tree .
```

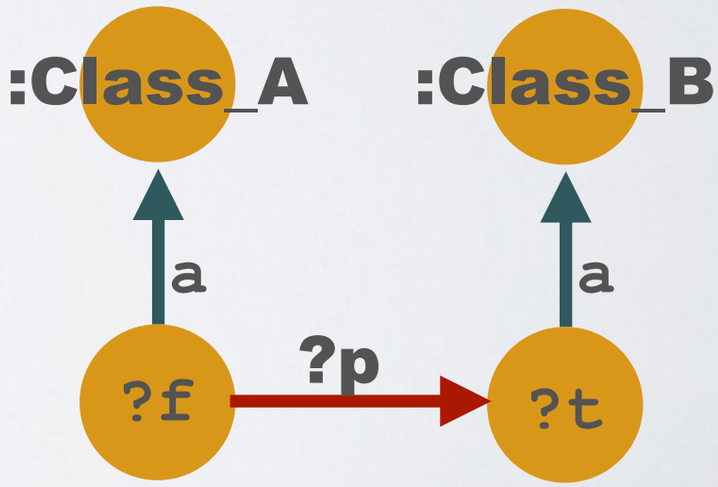
A locally undeclared class (LUC) instance

any resource that does not appear as an subject of **rdf:type** in the dataset.

Obtain metadata using SPARQL

Subject \ Object	Class A	Class B	LUC	Literal
Class A	0	76	23	50
Class B				
LUC				

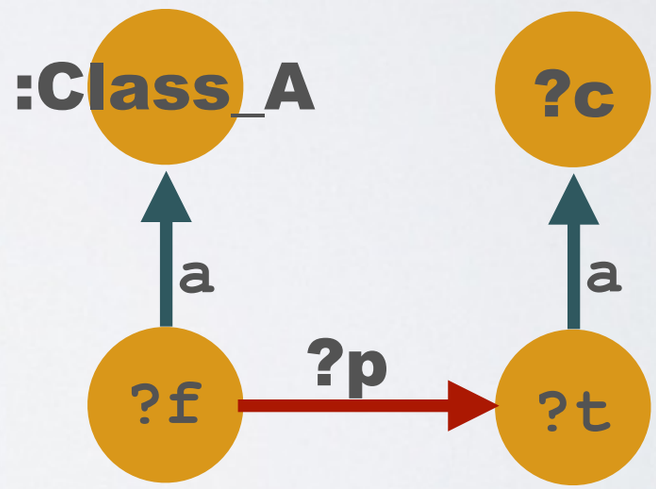
```
SELECT ?p (COUNT(?p) AS ?rc)
   WHERE {
      ?f ?p ?t .
      ?f a :Class_A .
      ?t a :Class_B .
   }
   GROUP BY ?p
```



Improvement

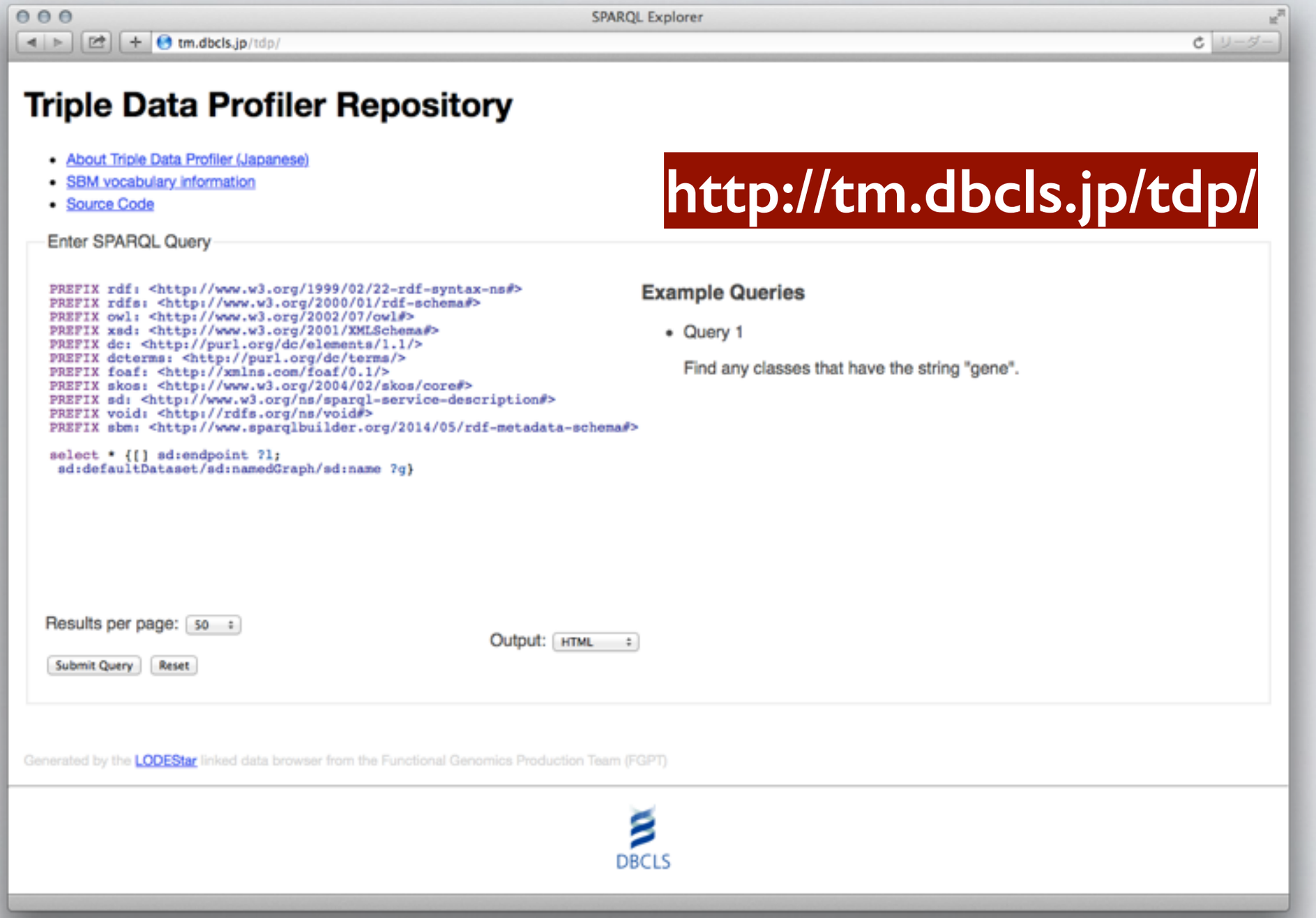
N^2 times (N: number of the classes)
Even a dataset of many classes has only a few relationships.
→ inefficient

```
SELECT ?p ?c (COUNT(?p) AS ?pc)
   WHERE {
      ?f a :Class_A ;
         ?p ?t ;
         !rdf:type ?t .
      ?t a ?c .
   }
   GROUP BY ?p ?c
```



→ N times

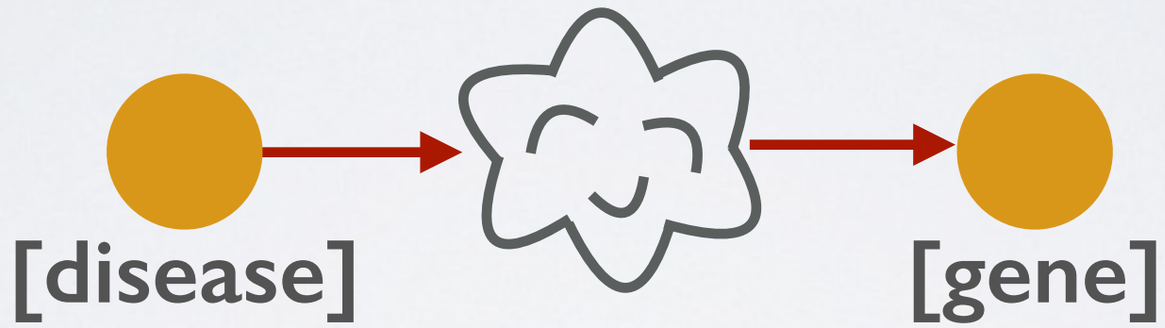
Gathering and providing metadata



Searching example

[gene] Class = bio2rdf:omim_vocabulary:Gene

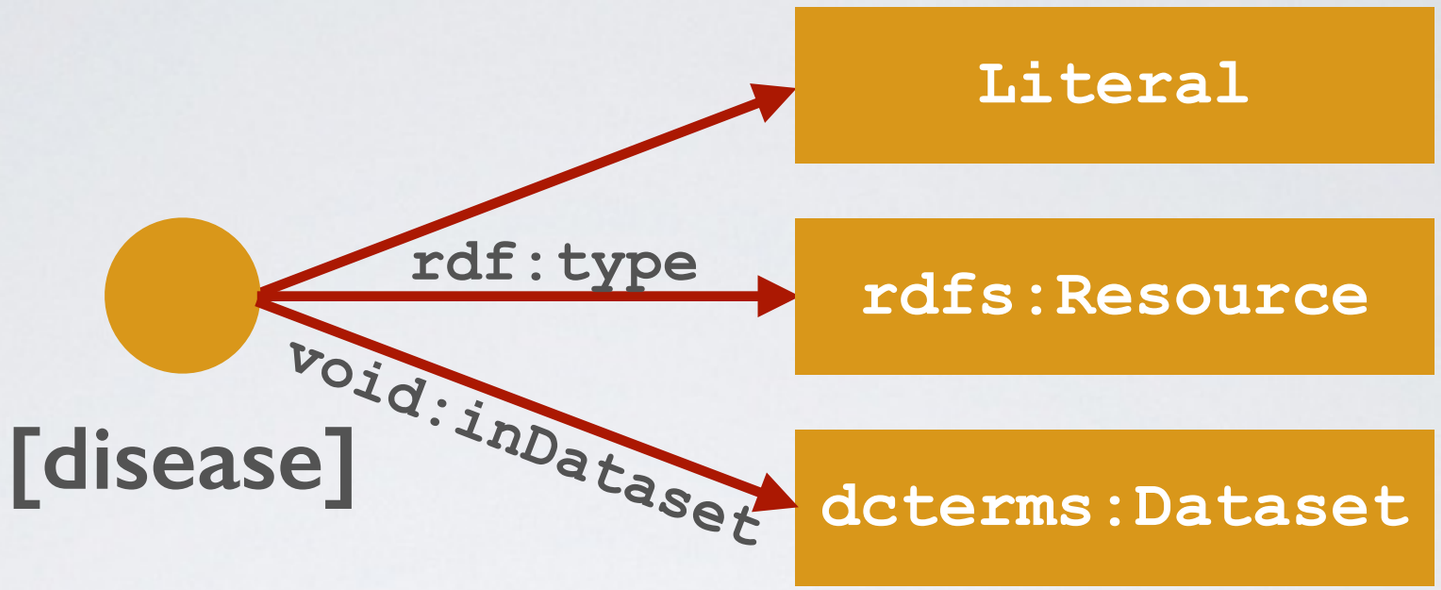
[disease] Class = bio2rdf:umls_vocabulary:Resource



OR

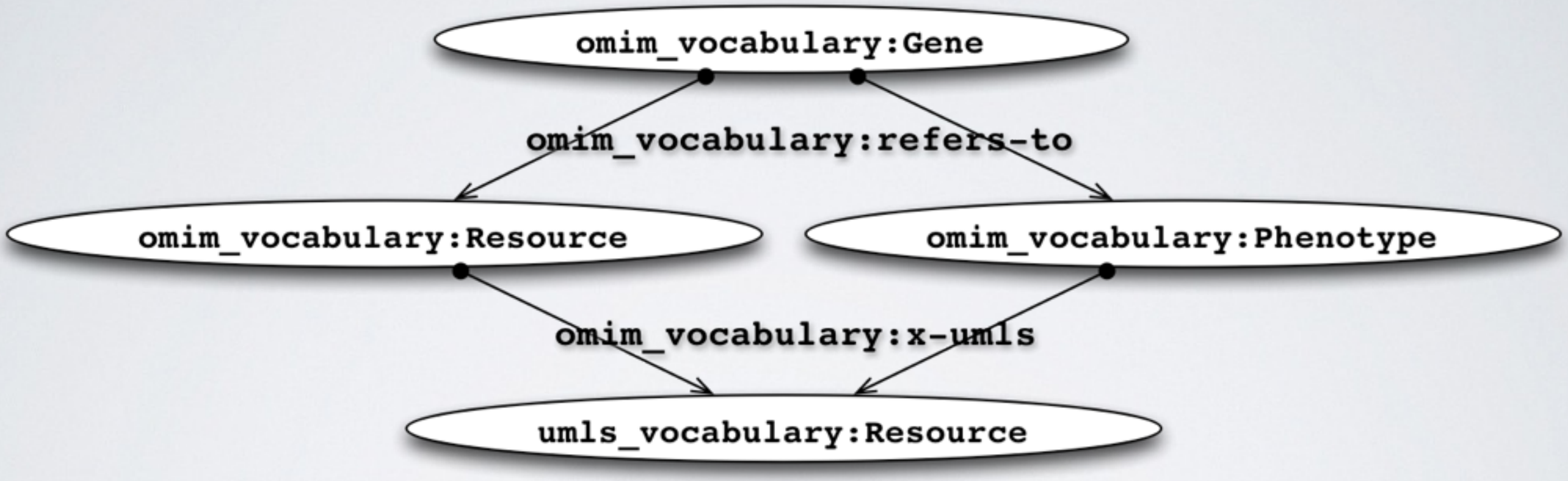


From [disease]



From [gene]

Count	Predicate	Object class
608220	bio2rdf-omim:x-gi	bio2rdf-gi:Resource
98639	bio2rdf-omim:article	bio2rdf-pubmed:Resource
95443	bio2rdf-omim:refers-to	bio2rdf-omim:Resource
81438	bio2rdf-omim:vocabulary:refers-to	bio2rdf-omim:Gene
28032	bio2rdf-omim:vocabulary:gene-symbol	bio2rdf-hgnc:Resource


omim_vocabulary:Gene
omim_vocabulary:refers-to
omim_vocabulary:Resource
omim_vocabulary:Phenotype
omim_vocabulary:x-umls
umls_vocabulary:Resource

Things to do

To improve inter-operability

Integrating vocabularies with other related parties.

To improve efficiency

Calling for data providers to provide metadata.

To realize a QA system using metadata

Discussing with LODQA developers.

Thank you!